Skin Cancer Detection System Using Deep Learning

Comprehensive Software Engineering Architecture, Code Breakdown & Dataset Guide

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DOCUMENT EXECUTIVE SUMMARY:
This document presents the complete technical specification of the Skin Cancer Detection web application. It explains: (1) what code resides in every file and its software engineering purpose, (2) the exact provenance of the HAM10000 / ISIC dermoscopy dataset, (3) how the dataset was ingested, preprocessed, augmented, and trained using MobileNetV2 transfer learning, and (4) the specific attributes, diagnostic categories, and benchmark cases extracted from the dataset.

1. Code Architecture & File-by-File Breakdown

The application follows the industry-standard **Model-View-Controller (MVC)** software pattern using Python Flask. Separating application configuration, persistence (SQLite), business logic (TensorFlow/ReportLab), and views (Jinja2) ensures high maintainability, testability, and strict patient data security.

File / Module	Contents & Implementation Details	Software Engineering Purpose
app.py	Flask application factory (create_app()), directory verification, blueprint registration, error handlers (400, 403, 404, 413, 500), and context injection.	Main entry point for local execution and production deployment.
config.py	Central configuration class: upload file size limits (10 MB), allowed formats (.jpg, .jpeg, .png), absolute paths for SQLite DB, uploads, and model weights.	Prevents hardcoding across modules; isolates runtime settings.
database/db.py	SQLite database engine (get_db_connection()), schema creation (users, patient_medical_profiles, analysis_records, audit_logs), automated non-destructive schema migrations, queries, and audit log writers.	Primary backend persistence layer. Ensures zero data loss across server restarts.
services/ auth_service.py	Password hashing with scrypt (salt + iterations), authentication verification, session management, and route authorization decorators (@login_required, @admin_required).	Guarantees Role-Based Access Control (RBAC) and prevents plain-text password storage.
services/ model_service.py	Image inspection (Pillow dimensions, channels, corruption check), normalization to [0.0, 1.0], TensorFlow MobileNetV2 inference (model.predict()), 7-class softmax probability scoring, and transparent fallback simulation.	Encapsulates computer vision and deep learning execution in an isolated service.
services/ pdf_service.py	ReportLab PDF generation: embeds lesion thumbnail, patient demographics, clinical classification, confidence gauge, medical disclaimers, and hospital-grade header/footer layout. Saves to uploads/reports/.	Produces formal clinical screening documents for patients and consulting dermatologists.

blueprints/ auth_bp.py	Routes for /login, /register, /logout, and /profile. Validates input credentials, manages sessions, and supports login by either email or username.	Handles user account lifecycle and profile updates.
blueprints/ analysis_bp.py	Routes for /upload, /analyze, /dashboard, /history, /results/<id>, /report/<id>, and /patient/medical-profile/update. Enforces patient data isolation (HTTP 403 for unauthorized cross-patient access).	Core patient interaction controller for lesion evaluation, symptom capture, and report downloading.
blueprints/ admin_bp.py	Routes for /admin (telemetry, KPI metrics, patient search), /admin/patient/<id> (clinical updates, physician notes, verification), and /admin/export-patients-csv with CWE-1236 formula injection defense.	Clinical command center for administrators and consulting dermatologists.
blueprints/ main_bp.py	Public informational routes: / (landing hero), /how-it-works (ML pipeline explanation), and /skin-health (educational ABCDE melanoma rule).	Educational awareness and public health screening outreach.
templates/ dashboard.html	Patient dashboard displaying personal medical card, active skin concerns, dermatological risk factors, physician notes, past scan archive, and clear actual vs. demo record badges.	Patient-facing view reflecting live synchronized backend records.
templates/admin/ patient_detail.html	Administrative patient detail view: medical history on the left, uploaded lesion photo gallery on the right, authorized verification dropdown, clinical notes form, and immutable audit trail.	Two-way synchronization interface between clinic and patient.
tests/ test_patient_sync.py	Automated Pytest test suite containing 12 comprehensive scenarios: registration persistence, server restart simulation, admin sync, cross-patient isolation (403), CSV formula defense, and schema idempotency.	Quality assurance suite guaranteeing 100% test pass rate across requirements.

2. Dataset Origin & Provenance

The model and educational benchmarks are powered by the **HAM10000 ("Human Against Machine with 10,000 training images")** dataset, recognized internationally as the benchmark research corpus for automated dermatological screening:

Institutional Authors: ViDIR Group (Department of Dermatology, Medical University of Vienna, Austria) and Cliff Rosendahl (School of Medicine, The University of Queensland, Australia).

Hosting Organization: The International Skin Imaging Collaboration (**ISIC Archive**), supported by the Society for Melanoma Research.

Dataset Acquisition Pipeline (download_ham10000.py):

- **Metadata CSV:** The official HAM10000_metadata.csv (containing 10,015 clinical records) is downloaded from verified academic repositories.
- **Dermoscopic Imagery:** High-resolution dermoscopic image files (.jpg) are retrieved directly from the official ISIC Archive Amazon Web Services S3 cloud repository: https://isic-archive.s3.amazonaws.com/images/{image_id}.jpg.
- **Integrity Verification:** Every downloaded photograph is programmatically opened and verified using Pillow (PIL) to eliminate truncated, corrupted, or non-RGB files prior to training.

3. Connection, Preprocessing & Deep Learning Training Pipeline

The model training workflow is implemented in train_model.py using TensorFlow and Keras. The end-to-end pipeline consists of four rigorous engineering stages:

Pipeline Stage	Technical Implementation	Clinical / Algorithmic Rationale
1. Lesion-Grouped Splitting	Dataset split into **Train (70%)**, **Validation (15%)**, and **Held-out Test (15%)** strictly grouped by lesion_id using Scikit-Learn.	**Eliminates Data Leakage:** Multiple photos of the same mole never appear across both train and test splits, preventing artificial over-optimism.
2. Input Preprocessing & Augmentation	Images resized to 224x224 RGB pixels. Normalized from [0, 255] to [0.0, 1.0]. On training images only, applied random horizontal/vertical flips, rotations (±15°), zoom (±10%), and contrast adjustments.	Standardizes tensor inputs for MobileNetV2 and teaches the CNN to recognize skin lesions regardless of patient orientation or ambient lighting.
3. Transfer Learning Architecture	**Base:** MobileNetV2 pretrained on ImageNet (1.4 million images). **Custom Top:** GlobalAveragePooling2D() → BatchNormalization() → Dense(128, ReLU) → Dropout(0.3) → Dense(7, Softmax).	Leverages general feature extractors (edges, cellular textures) while fine-tuning specialized top layers for dermatological lesion patterns.
4. Class Imbalance Compensation	Calculated balanced class weights: compute_class_weight('balanced', classes=..., y=...). Compiled with Adam(learning_rate=1e-4) and SparseCategoricalCrossentropy.	**Protects High-Risk Classes:** Benign moles represent ~67% of records. Class weighting forces the optimizer to treat rare/lethal melanomas with equal importance.

4. Specific Dataset Attributes & Reference Benchmarks Utilized

From the raw HAM10000 dataset, the application specifically extracts and utilizes four key data elements:

Code	Full Diagnostic Disease Category	Clinical Severity & Risk Level	Educational Reference Case Preloaded
mel	**Melanoma**	**Malignant Skin Cancer (High Risk / Urgent)**	HAM10000_ISIC_0029314.jpg (Superficial Melanoma)
bcc	**Basal Cell Carcinoma**	**Malignant Non-Melanoma**	HAM10000_ISIC_0024310.jpg (BCC Lesion)
akiec	**Actinic Keratoses / Intraepithelial Carcinoma**	**Pre-cancerous / Squamous Risk**	Preloaded in HAM10000 Benchmark Suite
nv	**Melanocytic Nevi**	**Common Benign Mole**	HAM10000_ISIC_0024306.jpg (Benign Nevus)
bkl	**Benign Keratosis-like Lesions**	**Non-Cancerous (Seborrheic Keratosis)**	HAM10000_ISIC_0024312.jpg (Solar Lentigo)
df	**Dermatofibroma**	**Benign Dermal Nodule**	Preloaded in HAM10000 Benchmark Suite
vasc	**Vascular Lesions**	**Benign (Angioma / Hemangioma)**	Preloaded in HAM10000 Benchmark Suite

1. Diagnostic Labels (dx column): Used as target ground-truth classification labels across the 7 disease classes shown above.

2. Image Identifier (image_id column): Used to index physical dermoscopic files (e.g. ISIC_0024306.jpg) and map them to histopathologic biopsy findings.

3. Lesion Identifier (lesion_id column): Used during data splitting to prevent cross-contamination and guarantee valid statistical evaluation.

4. Diagnostic Validation Type (dx_type column): Filters for histopathologically verified ground-truth lesions (gold-standard biopsies).

5. Preloaded Educational Benchmark Cases: Embedded into pre-registered demo accounts (**Bhavi Doshi** and **Vidisha Ajmera**) to allow lab examiners, viva evaluators, and students to test full image inspection, risk categorization, and PDF generation workflows reliably without requiring internet connectivity.

ACADEMIC & CLINICAL DISCLAIMER:
This document and the associated software application are developed for educational evaluation and preliminary screening demonstration at SVKM's NMIMS MPSTME. This software is NOT a certified clinical diagnostic medical device. Artificial intelligence algorithms can produce false positive and false negative results. Always consult a licensed dermatologist or medical oncologist for clinical dermoscopy, biopsy, and professional healthcare advice.